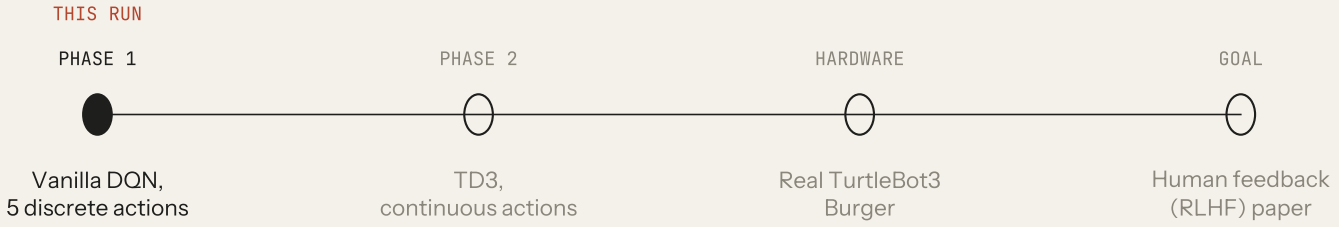
Closing the gap

between a fast 2D simulator and Gazebo, for a robot learning to reach goals



Where this run sits

Phase 1 trains a standard DQN agent in two simulators that share one interface: a pure-Python 2D simulator that is fast enough for full training, and Gazebo, which is slower but closer to the real robot. A policy is trained in 2D and then tested in Gazebo. The previous report found that policies lost 23 points of success in that move, without knowing why. This run had three jobs:

A Explain the 23-point drop, then fix it or show it can't be fixed.

B Report final numbers on layouts no decision ever used, with 10 seeds instead of 3.

C Show training is stable and that results reproduce exactly.

23 to 1.6

points of success lost moving to Gazebo

73.7%

goal success, 10 seeds, 200 held-out layouts

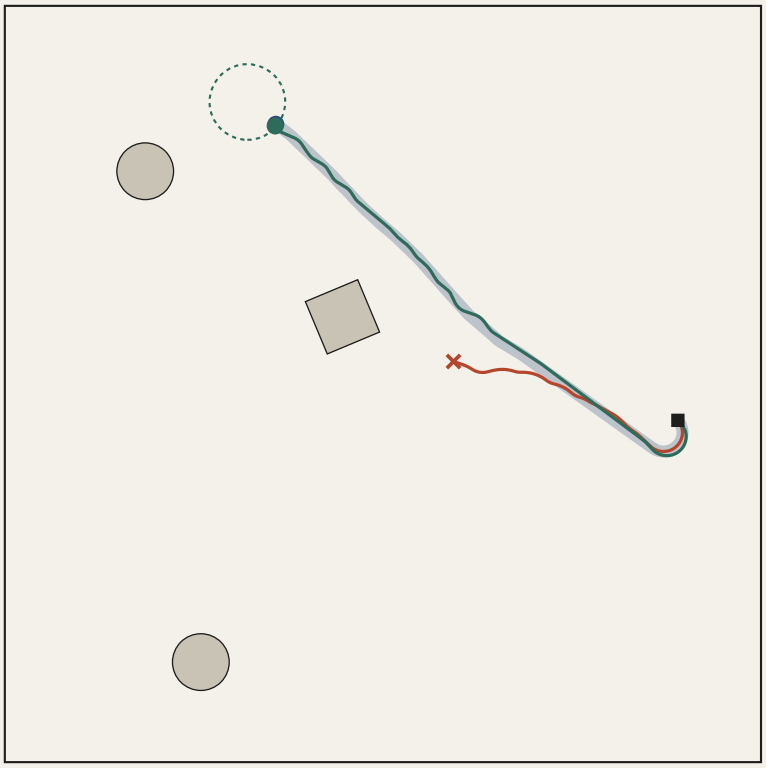
bit-exact

same seed reruns and resumed runs match

Vanilla DQN only (no Double, Dueling or prioritised replay). Every number in this document comes from files in results/.

The robot believed it was elsewhere

Same policy, same layout, three simulators. The 2D path is the broad pale band.



2D simulator

Gazebo, before the fix

Gazebo, after the fix

layout 2 · seed 0 · square start · ring goal · x where Gazebo's LiDAR read a collision while the env, trusting wheel odometry, placed the robot in open floor

Two defects in the Gazebo tier

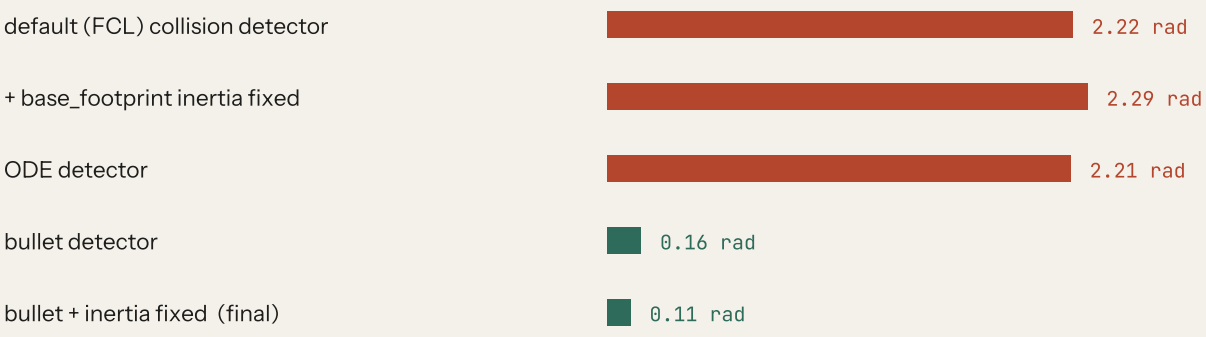
[1] **Wheel grip**

After a held hard turn, Gazebo's default collision detector let the body keep spinning for 1–2 s while the wheels drove straight.

[2] **Wrong pose**

The env read its position from wheel odometry, which cannot see that spin. Heading to the goal was off by up to 2 rad.

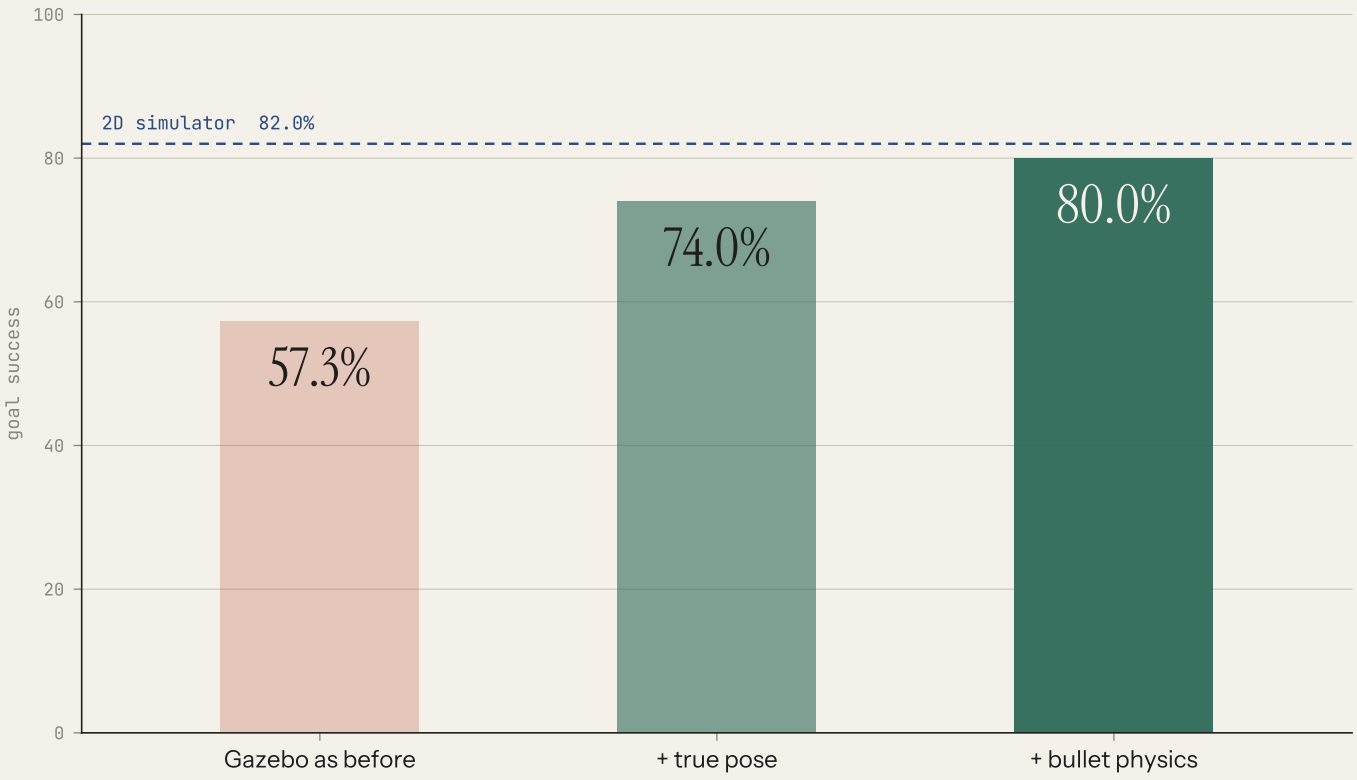
Uncommanded yaw after a hard turn, measured against the physics engine's true pose



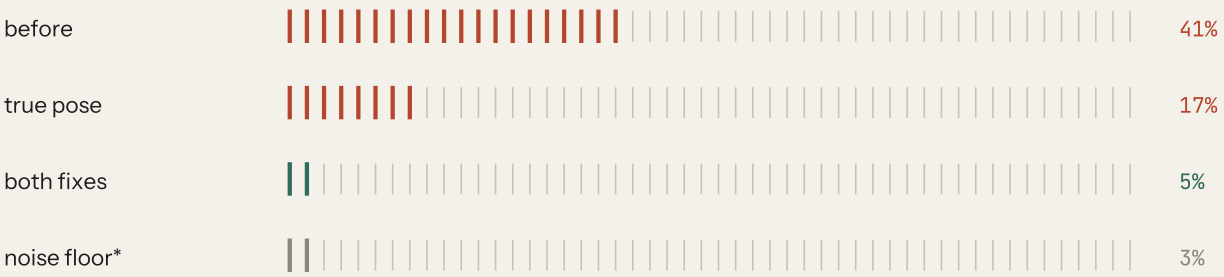
Only 7% of the goals Gazebo recorded before the fix were real: scan-matching the LiDAR against the known layout put the robot within 0.2 m of the goal in 6 of 86 'successes'. Fix: bullet detector, true pose from the physics engine.

One fix at a time

Same three checkpoints, same 50 layouts. Only the Gazebo tier changes.



Outcome flips: layouts where the two simulators disagree on goal vs collision

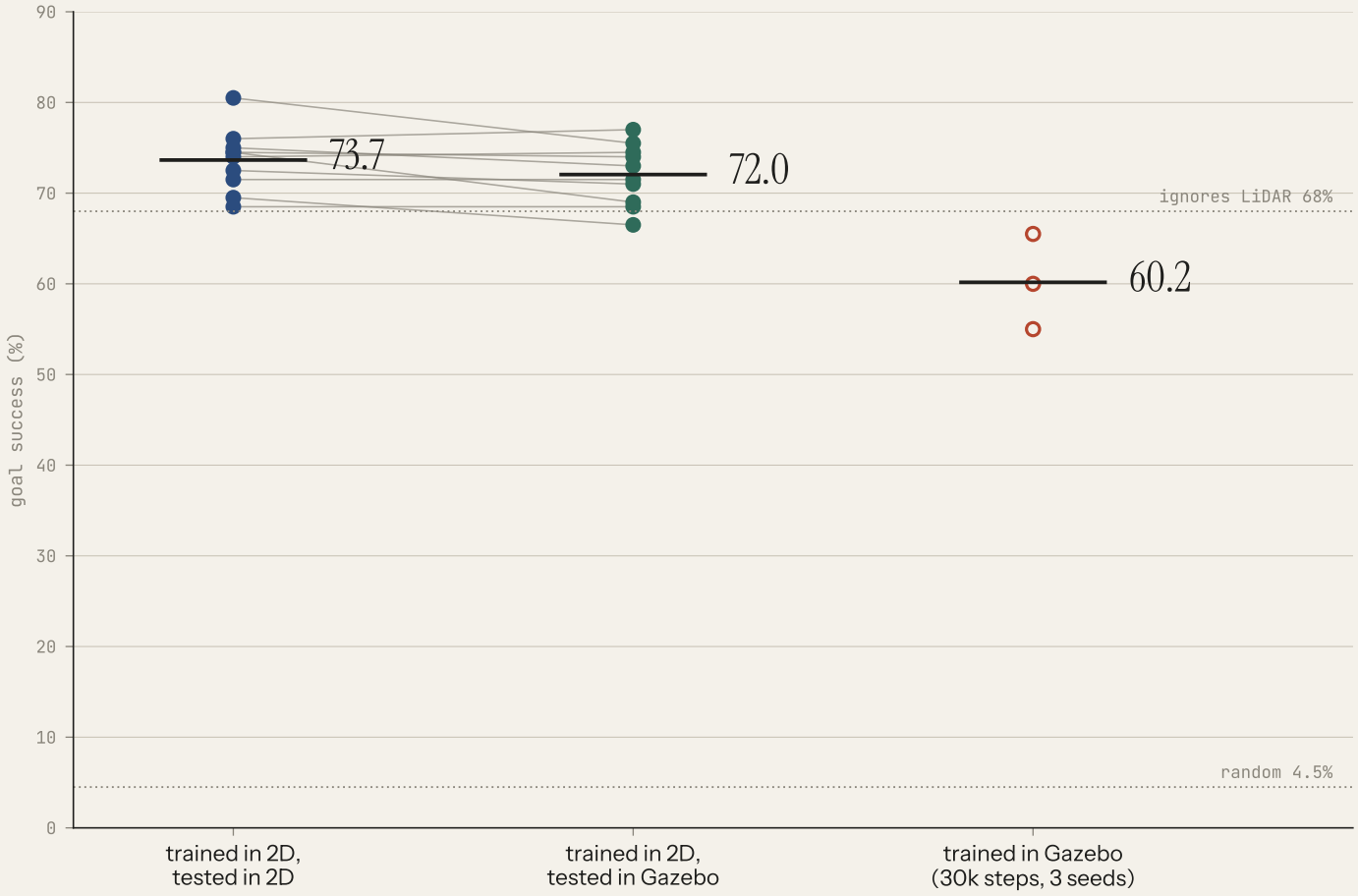


* the 2D simulator against itself with a different LiDAR-noise seed: the policy's own chance level.

Ruled out: the 47–56 ms command delay Gazebo adds. Injecting 0–90 ms into the 2D simulator left success at 81–83%, and retraining with it moved Gazebo success by +3.3 points, which is noise. Collision rule and LiDAR geometry already match.

Ten seeds, two simulators

Last checkpoint of every run, 200 layouts that no decision ever used.

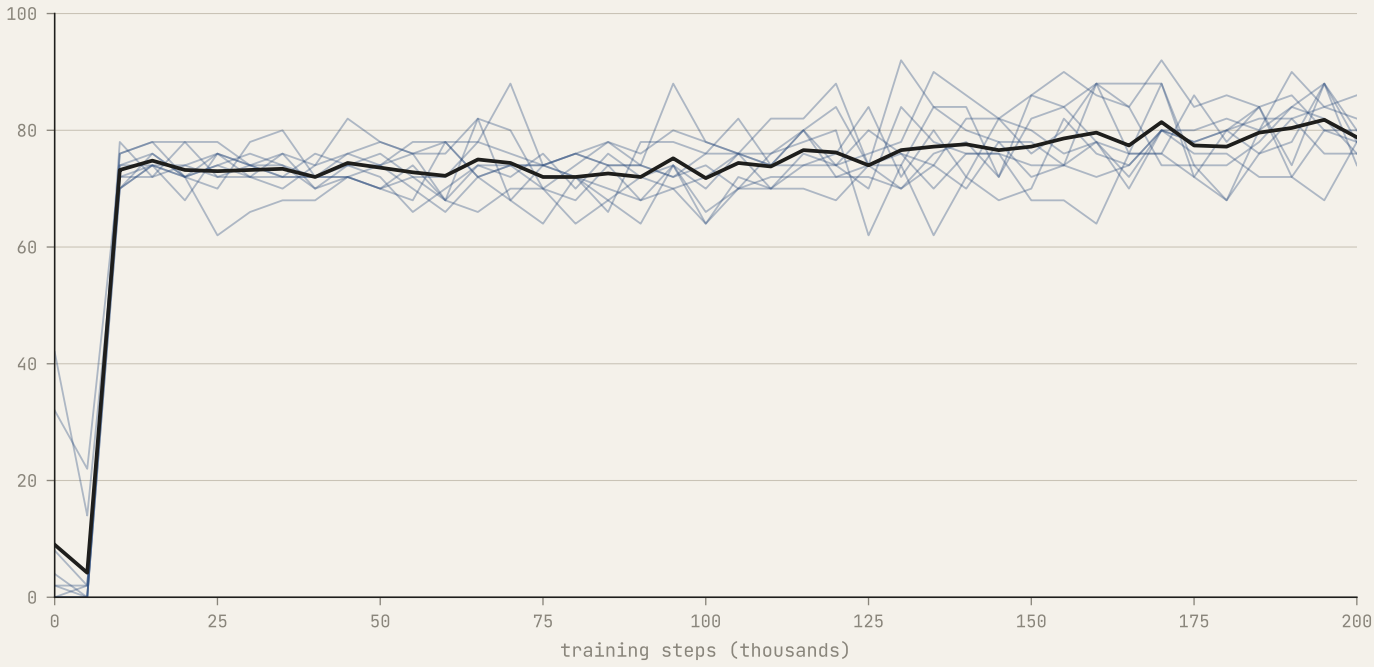


CONDITION	SEEDS	SUCCESS %	COLLISION %
Trained in 2D, tested in 2D	10	73.7 ± 2.4	26.4
Trained in 2D, tested in Gazebo	10	72.0 ± 2.4	27.9
Trained in Gazebo, 30k steps	3	60.2 ± 13.0	39.8
Heading only, ignores LiDAR	-	68.0	32.0
Random actions	-	4.5	95.5

Transfer gap 1.6 points (95% CI 0.0 to 3.2), paired over seeds; 5.9% of layout outcomes differ. Failures are collisions; timeouts are 0%. DQN beats a controller that ignores obstacles by about 6 points: a real but modest margin, and the headroom for phase 2.

Stable, and exactly repeatable

Success on the tuning layouts during training, every seed drawn.



Ledger

- **No collapse** — 16 shallow dips in 390 evaluations; deepest 20 points. The untuned run once fell to 3%.
- **No Q divergence** — Mean max-Q rises smoothly to ~70; loss falls slowly; Q at dips is no higher than elsewhere.
- **Mild overestimation** — Late in training Q(start) is about 53 vs a realised discounted return of 51.
- **Same seed, same run** — A rerun matches the original bit for bit, down to the final network weights.
- **Survives restarts** — Three runs cut by real WSL restarts resumed to bit-identical results.
- **Provenance** — Each run stores package versions and hashes of the config and every source file.

Caveats

One task, one reward, one robot model. Gazebo now uses the true pose, which a real robot will not have.
Gazebo is not deterministic, so its numbers repeat statistically (about 3 points per checkpoint), not bit for bit.
The Gazebo-trained baseline has 3 seeds and a 30k-step budget.